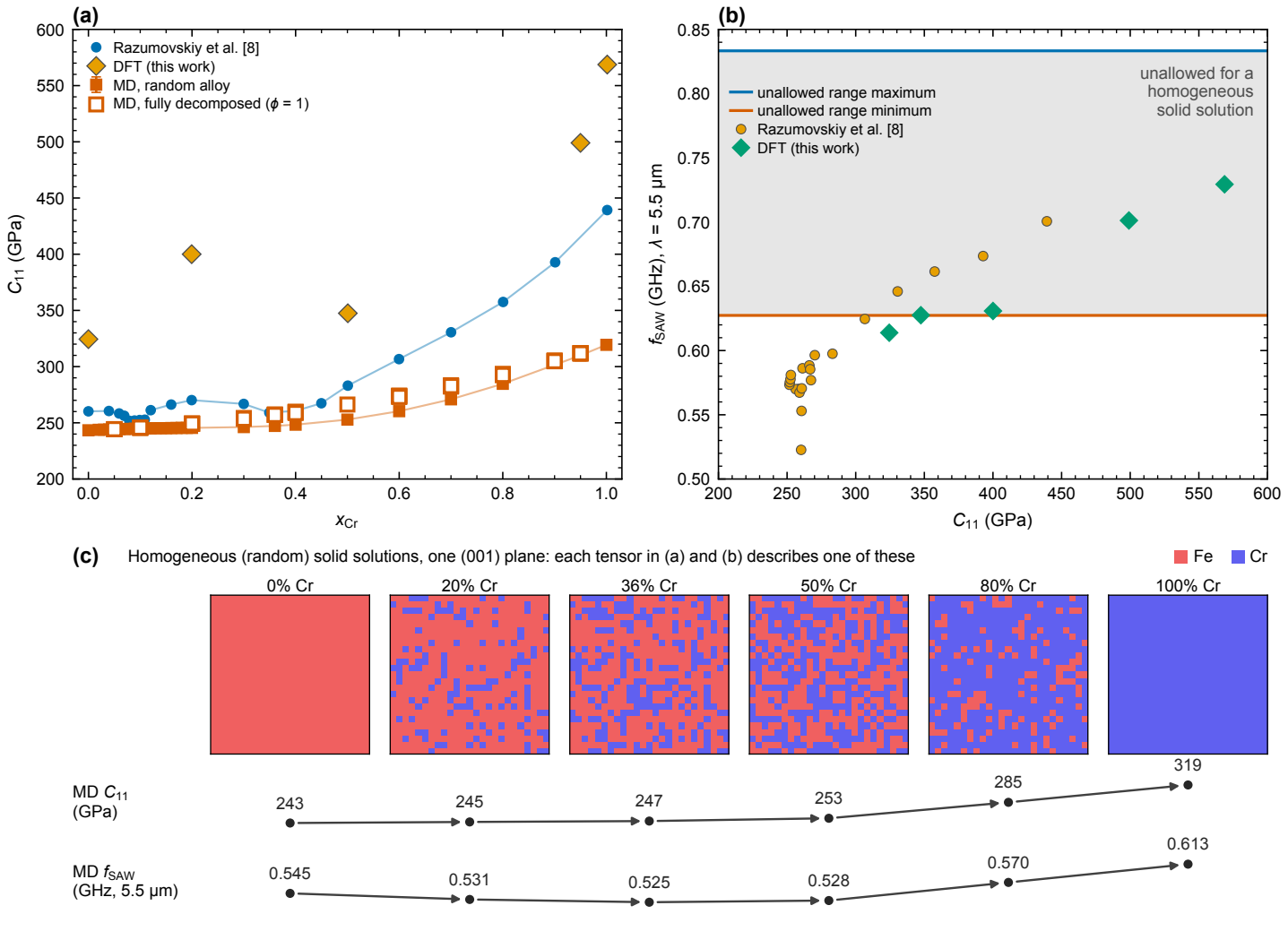

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
MD, fully decomposed (ϕ = 1)
C11 (GPa)
xCr
(b)
unallowed range maximum
unallowed range minimum
Razumovskiy et al. [8]
DFT (this work)
unallowed for a homogeneous solid solution
fSAW (GHz), λ = 5.5 μm
C11 (GPa)
(c)
Homogeneous (random) solid solutions, one (001) plane: each tensor in (a) and (b) describes one of these
Fe
Cr
0% Cr
20% Cr
36% Cr
50% Cr
80% Cr
100% Cr
MD C11 (GPa)
243
245
247
253
285
319
MD fSAW (GHz, 5.5 μm)
0.545
0.531
0.525
0.528
0.570
0.613